Discussion Questions

1: *True or false?*

• if the hydrostatic (gauge) pressure at a depth h in the ocean is 2.5 atmospheres, then at a depth of $2h$ the hydrostatic (gauge) pressure will be 5.0 atmospheres.

• if an object is floating, the buoyant force on it is equal to its weight.

• water flows through a pipe having a varying width. More water flows per second through the wide section than through the narrow section because there is more room for it to flow.

2 A jet of air (e.g. from a leaf blower) is directed between two balloons hanging a little way apart. Will the balloons move further apart, closer together, or be unaffected?

3 A jar of Neon gas sits on a bench next to a jar of Helium. Both jars have the same volume and their contents weigh the same. They are in equilibrium with the surroundings at 25°C. Rank the gases in order of

a) Molecular rms velocity
b) Average molecular kinetic energy
c) Pressure

4 Smaller planets like Mercury have a low escape velocity, which is the speed a particle needs to reach to escape the gravitational pull of the planet (kinetic energy > gravitational energy). Smaller planets also have very little to no atmosphere remaining. How can we use the kinetic theory of gases to explain this atmosphere loss?

Problem-solving Questions

5 The Tonga Trench in the Pacific Ocean is 10 km deep. Calculate the *absolute* pressure at the bottom of the trench in pascal. (DATA: seawater density = 1025 kg m^{-3}, assume constant)

6 The earth's atmosphere has a pressure versus altitude relation approximated by $P = P_0\exp(-\alpha h)$, where $\alpha = 0.116$ km^{-1}. As QF30 was flying from the UK to Australia at 10,000 m, the internal cabin pressure was equivalent to the atmosphere at 2500 m. Calculate the force exerted by air pressure on a square metre of cabin wall area. Take $P_0 = 101$ kPa.

7 A 120 cm^3 block of aluminium ($\rho = 2700$ kg m^{-3}) is hung fully immersed on the end of a piece of string in a bucket of water ($\rho = 1000$ kg m^{-3}). Take $g = 9.8$ N kg^{-1}.

• calculate the tension in the string

• the apparatus is now placed in a lift accelerating downwards at 1.8 m s^{-2}; does this effect the buoyancy force? If so, how? Either way, calculate the tension in the string now

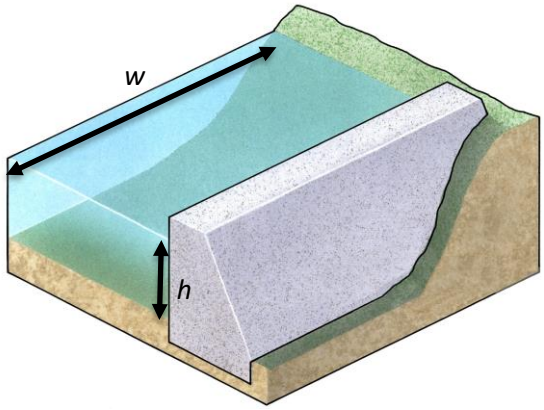
• the apparatus is now taken on an amusement park ride. In part of the ride the passengers are in a 'free fall' situation; calculate the tension in the string now

8 How much work is done by pressure in forcing 1.4 m^3 of water through a pipe with an internal diameter of 13 mm if the pressure difference at the two ends is 1.0 atmosphere?

9 What is the rms velocity of Argon gas at 40°C ($M_{Ar} \simeq 40$u)?

10 (mini-challenge question)

A dam wall of width *w* is holding back water of density 1000 kg.m^{-3} with a depth *h.* Calculate the total force exerted *by the water* on the dam wall. Take g = 9.8 m s^{-2}, h = 30m and w = 100m. You can assume that the density of water is 1000kg m^{-3}

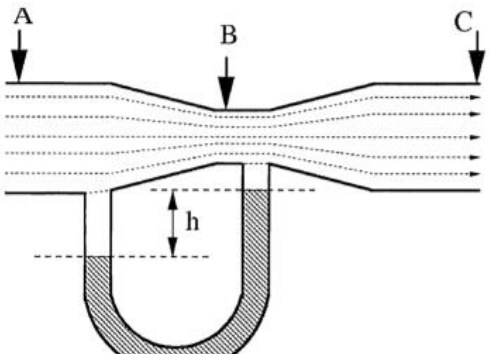


Past exam questions

11 A solid spherical object floats in water with half of its volume submerged.

(i) Given that the density of water is $\rho_w = 1000\text{kg.m}^{-3}$, what is the density of the object?
(ii) If the object is placed in paraffin oil of density $\rho_{parafin} = 800\ \text{kg.m}^{-3}$, what fraction of the sphere's volume will be submerged?

12 A venturi tube is shown in the diagram, with the flow direction marked by horizontal arrows. The cross-sectional areas of the fluid stream at points A, B and C are a m^2, a/2 m^2 and a m^2 respectively. The density of the incompressible fluid flowing in the tube past points A, B and C is ρ kg.m^{-3}.



a) If the speed of the fluid at point A is v ms^{-1}, what is the speed of the fluid at:
   i) Point B?
   ii) Point C?
   [Express your answers in terms of v.]

b) Find an expression for the pressure difference between points A & B in terms of ρ & v.
c) If the incompressible fluid in the curved section of tubing (indicated by the shading) has a density of σ kg m^{-3}, find an expression for v in terms of σ, ρ *and* g (the gravitational acceleration constant in ms^{-2}) and h (difference in the fluid levels in metres)

13 A cylinder of a monatomic gas has a volume of 15,000 cm^3 and a pressure of 100 atm.
   a) What is the thermal energy of this gas at room temperature (20°C)?
   b) What is the mean free path in the gas (r = 1.0 x 10^{-10} m)?

Short solutions

Discussion Questions:

1 True (unless you take into account the changing density of water); **true** (definition of floating); **false**(equation of continuity)

2 Closer together (follows from Bernoulli)

3 v_{rms}(He) > v_{rms}(Ne), K(He)=K(Ne), P(He)>P(Ne)

4 Speed distribution – anything above the escape velocity will leave the planet, and the remaining particles will redistribute their speeds such that some new particles have a speed greater than the escape speed, and they too will leave the planet. This process repeats until there is almost no atmosphere left.

Problem Solving Questions:

5 101 MPa

6 43.9 kN

7 2.0 N; 1.6 N; 0 N

8 1.4×10^5 J

9 442 m/s

10 441 MN

Past Exam Questions:

11 500 kg m^{-3}; 5/8

12a i. $2v$ **ii.** v **b** $\frac{3}{2}\rho v^2$(at mid point of flow) **c** $v = \sqrt{\frac{2\sigma g h}{3\rho}}$

13a $2.28\text{x}10^5$ J **b** $2.24\text{x}10^{-9}$ m